

Day 4 Lecture Notes: Linear Programming and Optimization

Mathematical Modeling & Computational Projects Camp

Morning Structure (10:00–12:30)

- 10:00–10:30: Lecture Block 1
- 10:35–11:05: Problem Solving Session 1 (30 minutes)
- 11:10–11:40: Lecture Block 2
- 11:40–12:10: Problem Solving Session 2 (shorter) + 12:10–12:30 synthesis/review

Lecture Block 1 (10:00–10:30): What Optimization Means in LP

Optimization means finding the best value (maximum or minimum) under rules.

In linear programming (LP), we have:

- Decision variables (what we choose),
- Linear constraints (limits),
- Linear objective (what to maximize/minimize).

Example form:

maximize $z = ax + by$ subject to linear constraints.

Vocabulary focus:

- **Feasible solution:** satisfies every constraint.
- **Feasible region:** all feasible solutions together.
- **Optimal solution:** feasible point with best objective value.

Instructor prompt: “Can a point have high profit but still be invalid?”

Mini example (model setup only):

- Variables: x = posters, y = flyers.
- Resource limits: printing time and design time.
- Objective: maximize revenue $R = 6x + 4y$.

Message to emphasize: optimization is about best *valid* choice, not just best number.

Problem Solving Session 1 (10:35–11:05): Practice Set A (30 minutes)

Goal: Translate context to LP components and test candidate points.

Suggested pacing:

- First 10 minutes: Problems 1–4.
 - Next 12 minutes: Problems 5–8.
 - Last 8 minutes: Problems 9–10 and group check.
1. Define decision variables for making posters and flyers.
 2. Write inequalities: $2x + y \leq 120$, $x + 3y \leq 150$, $x, y \geq 0$.
 3. Explain in words what each inequality means.
 4. State objective for maximizing $P = 5x + 4y$.
 5. List two feasible points and one non-feasible point.
 6. For points $(0, 0)$, $(20, 30)$, $(40, 10)$, compute $P = 5x + 4y$.
 7. Which tested point from Problem 6 is best?
 8. Why must nonnegativity constraints be included?
 9. Test whether $(30, 25)$ is feasible under the constraints in Problem 2.
 10. Write one realistic assumption that makes this LP linear.

Instructor checkpoint questions:

- “Which constraint is active at your best tested point?”
- “How would changing a coefficient in the objective affect your decision?”

Lecture Block 2 (11:10–11:40): Feasible Region and Corner Principle

The feasible region is the set of all points satisfying constraints. For 2-variable LPs, the best solution occurs at a corner point (vertex), when a best solution exists.

Worked example:

$$\begin{array}{ll}\text{maximize} & 3x + 2y \\ \text{subject to} & x + y \leq 100, \ 2x + y \leq 140, \ x, y \geq 0.\end{array}$$

Test corner points to identify best value.

Corner computation walkthrough:

- Corner 1: $(0, 0)$.
- Corner 2: $(70, 0)$ from $2x + y = 140$ with $y = 0$.

- Corner 3: $(0, 100)$ from $x + y = 100$ with $x = 0$.
- Corner 4: intersection of $x + y = 100$ and $2x + y = 140$: subtract to get $x = 40$, then $y = 60$.

Objective check:

$$z(0, 0) = 0, \quad z(70, 0) = 210, \quad z(0, 100) = 200, \quad z(40, 60) = 240.$$

Best corner is $(40, 60)$ with $z = 240$.

Interpretation: the optimum occurs where constraints trade off most efficiently.

Problem Solving Session 2 (11:40–12:10): Practice Set B (shorter)

Goal: Use corner-point method with interpretation.

Suggested pacing:

- First 12 minutes: Problems 1–4.
 - Next 10 minutes: Problems 5–7.
 - Last 8 minutes: Problems 8–9 and explanation write-up.
1. Graph: $x + y \leq 12$, $x \leq 8$, $y \leq 7$, $x, y \geq 0$.
 2. List all corner points of the region.
 3. Compute $z = 5x + 4y$ at those corners.
 4. State maximizing corner and objective value.
 5. Give one real-world interpretation of your result.
 6. Add new constraint $x \leq 6$. Recompute candidate corners that change.
 7. Under the new constraint in Problem 6, does the optimum change? Explain.
 8. Write one sentence describing what an active (binding) constraint means at your optimal point.
 9. Reflection: give one reason a graph-based LP method is less practical for many variables.

Instructor move for fast finishers: ask for a minimization version of the same model and a comparison of where the optimal point tends to sit.

Synthesis and Review (12:10–12:30)

- Revisit: variable, constraint, feasible, objective, corner.
- Class check of one full LP from words to answer.
- Exit check: “Why do we test corners?”
- Exit check 2: “Can the highest objective value occur at a non-feasible point? Why not?”
- Exit check 3: “What happens when we add a new constraint?”

Python Preview (Afternoon)

Use `scipy.optimize.linprog` to solve LP models and compare with graph-based answers.